

Paper I: Finite-Horizon Value Analysis for a Simple Markov Reward Process

April 30, 2026

Abstract

This note summarizes a first set of finite-horizon Markov Reward Process (MRP) experiments for a four-state example with transition-level rewards. For a grid of discount factors $\gamma \in \{0, 0.05, \dots, 1\}$, we compute exact finite-horizon state values by Bellman recursion and compare them against Monte Carlo estimates with confidence intervals. We report two main runs, both using 100 trajectories per state and discount factor, with horizons $H = 20$ and $H = 50$. The exact ranking of states is stable across the full grid and both horizons: **Computer** > **Coffee** > **Home** > **Chat**.

1 The MRP

The state space is

$$S = \{\text{Chat}, \text{Coffee}, \text{Computer}, \text{Home}\}.$$

The transition probabilities and rewards are:

From	To	Probability	Reward
Chat	Chat	0.5	-1
Chat	Coffee	0.2	1
Chat	Computer	0.3	2
Coffee	Chat	0.7	2
Coffee	Coffee	0.1	1
Coffee	Computer	0.2	3
Computer	Chat	0.1	-3
Computer	Coffee	0.2	1
Computer	Computer	0.5	5
Computer	Home	0.2	2
Home	Coffee	0.4	1
Home	Home	0.6	1

Figure 1 shows the visualization used in this project.

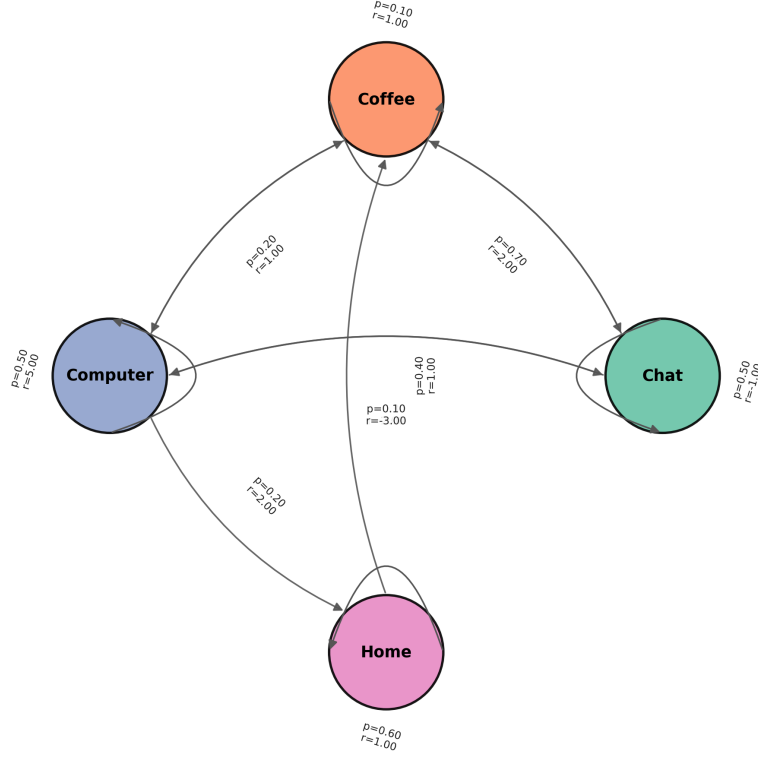


Figure 1: The four-state Markov Reward Process. Edge labels show transition probability p and immediate reward r .

2 Exact Finite-Horizon Values

For a horizon H , define the finite-horizon return

$$G_0^{(H)} = r_1 + \gamma r_2 + \dots + \gamma^{H-1} r_H.$$

The exact finite-horizon value function is

$$V_H(s) = \mathbb{E}[G_0^{(H)} \mid s_0 = s].$$

Because rewards are attached to transitions, the Bellman recursion is

$$V_0(s) = 0,$$

$$V_H(s) = \sum_{s'} P(s' \mid s) (R(s, s') + \gamma V_{H-1}(s')).$$

This recursion is exact and cheap for the present MRP, so it serves as the reference quantity for all comparisons against Monte Carlo estimates.

3 Experimental Protocol

For each $\gamma \in \{0, 0.05, \dots, 1\}$ and each start state, we ran:

- exact finite-horizon evaluation by Bellman recursion;
- Monte Carlo estimation from 100 trajectories;
- a 95% confidence interval for the Monte Carlo mean.

Two main result directories were used:

- `MRP/outputs_100traj_th10`, whose CSV files actually record $H = 20$;
- `MRP/outputs_100traj_th50`, with $H = 50$.

The directory name `th10` is therefore misleading; the paper uses the actual horizon value stored in the output files.

4 Main Results

4.1 State ranking

Across the full γ grid and both horizons, the exact ordering never changes:

`Computer > Coffee > Home > Chat.`

This is already visible at $\gamma = 0$, where values reduce to expected one-step rewards, and it remains true up to $\gamma = 1$ under the chosen finite horizons.

4.2 Representative exact values

Table 1 lists exact values at three representative discount factors.

Horizon	γ	<code>Computer</code>	<code>Coffee</code>	<code>Home</code>	<code>Chat</code>
20	0.00	2.8000	2.1000	1.0000	0.3000
20	0.50	4.6158	3.3484	2.3852	1.7696
20	1.00	32.1830	30.3916	29.1096	29.0409
50	0.00	2.8000	2.1000	1.0000	0.3000
50	0.50	4.6158	3.3484	2.3852	1.7696
50	1.00	77.5669	75.7755	74.4935	74.4248

Table 1: Representative exact finite-horizon values. Increasing the horizon has no visible effect at $\gamma = 0$, little effect at moderate discounting, and a large effect when $\gamma = 1$.

4.3 Monte Carlo accuracy

Table 2 summarizes absolute errors between the Monte Carlo estimates and exact values over the full state-discount grid.

Horizon	Mean absolute error	Maximum absolute error	Worst- γ mean error
20	0.3016	2.4670	$\gamma = 1.00$: 1.7840
50	0.2500	2.3252	$\gamma = 1.00$: 1.1398

Table 2: Aggregate Monte Carlo error summary using 100 trajectories per state and discount factor. The hardest regime is $\gamma = 1$, where long-horizon variance is highest.

The finite-horizon Monte Carlo estimates are generally close to the exact values for small and moderate γ , but the error grows substantially near $\gamma = 1$. This is expected: as discounting weakens, later rewards matter more, and finite-sample variance becomes larger.

5 Figures

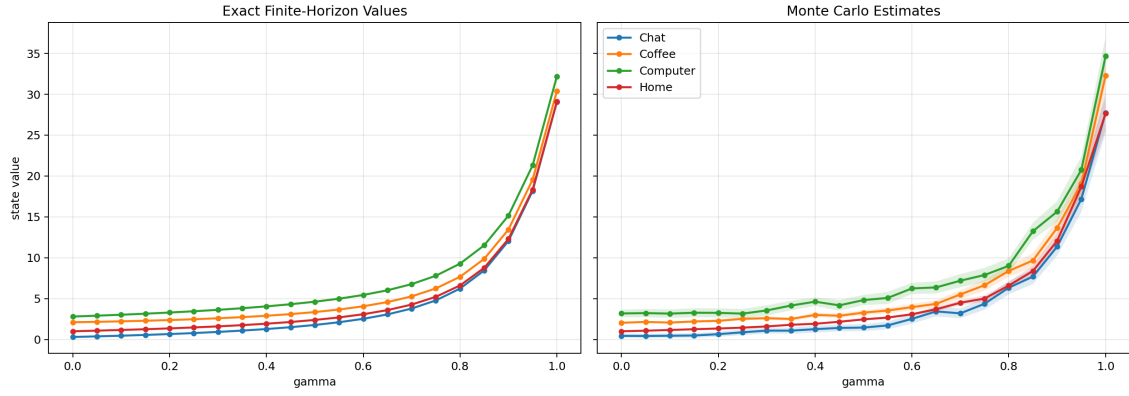


Figure 2: Exact values and Monte Carlo estimates with 95% confidence bands for the run stored in `outputs_100traj_th10`. The CSV metadata show that this corresponds to horizon $H = 20$.

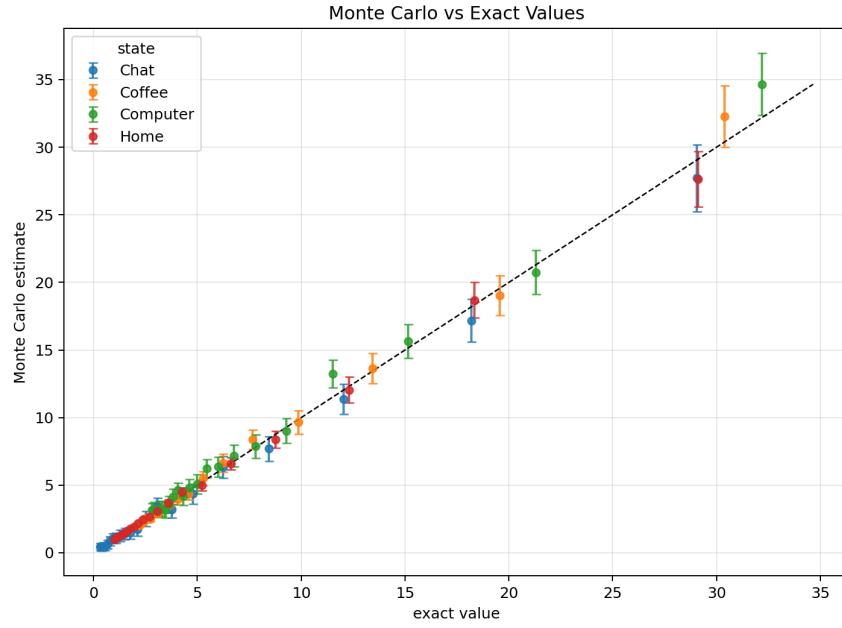


Figure 3: Exact vs Monte Carlo estimates for the $H = 20$ run. The diagonal line is the identity reference.

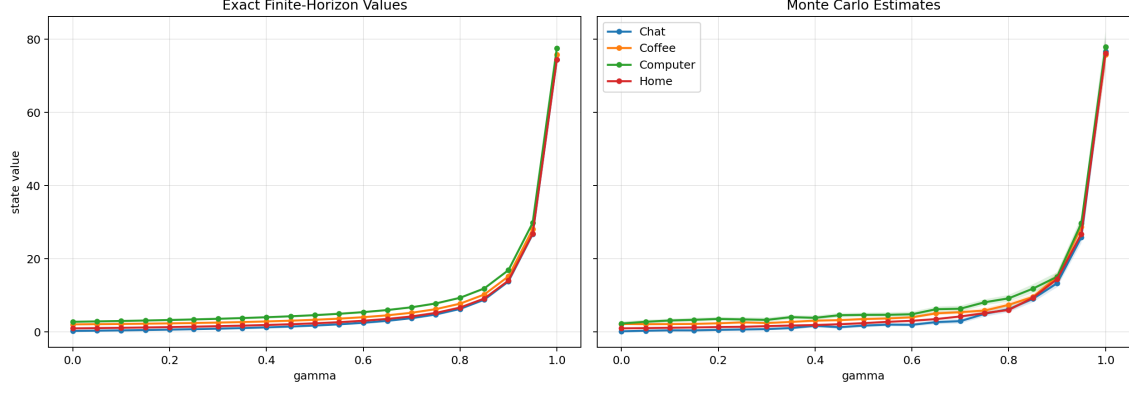


Figure 4: Exact values and Monte Carlo estimates with 95% confidence bands for horizon $H = 50$.

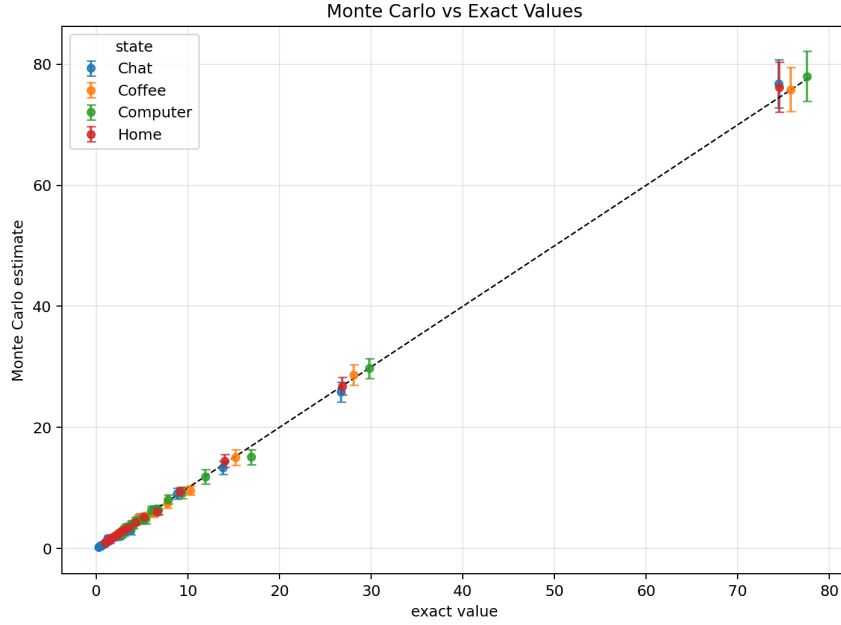


Figure 5: Exact vs Monte Carlo estimates for the $H = 50$ run.

6 Interpretation

Three simple points stand out.

1. **The ranking is robust.** Computer is always the highest-value start state, followed by Coffee, then Home, then Chat.
2. **Horizon matters most when γ is large.** At $\gamma = 0.5$, the exact values for $H = 20$ and $H = 50$ are effectively identical at the reported precision. At $\gamma = 1$, the longer horizon produces much larger values.
3. **Monte Carlo variance is the limiting factor near $\gamma = 1$.** Even with 100 trajectories, the difficult region is the weak-discount regime, where the cumulative reward has higher variance.

7 Conclusion

This first paper establishes a clean baseline for the project: the simple MRP is fully specified, exact finite-horizon values are available by Bellman recursion, Monte Carlo estimates can be checked against ground truth, and the main qualitative ranking is stable across horizons and discount factors. The most natural next extensions are infinite-horizon analysis for $\gamma < 1$, trajectory-count sensitivity, and ensemble experiments on randomly generated MRPs.